

Chapter 11. Airway Pressure Release Ventilation

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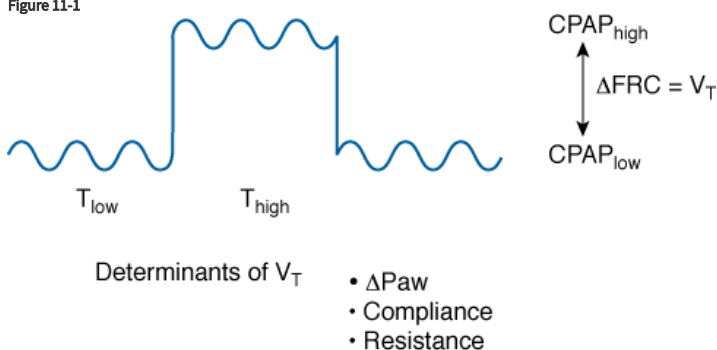
Airway Pressure Release Ventilation: Introduction

Controlled mechanical ventilation (CMV) is traditionally provided via an artificial airway to completely unload a patient's work of breathing and assure adequate gas exchange during the acute phase of respiratory insufficiency, until the underlying respiratory function has resolved.¹ The criteria used to determine when to terminate mechanical ventilation are essentially based on the clinical, and often, subjective assessment of the intensive care physician or on standardized weaning methods.^{2,3} The actual process of weaning the patient from CMV is carried out by allowing spontaneous breathing attempts with a T piece or continuous positive airway pressure (CPAP) or by gradually reducing mechanical assistance.^{4,5} Not surprisingly, gradual reduction of partial ventilator support benefits only patients who have difficulty in sustaining unassisted breathing.⁴ Although introduced as weaning techniques, partial support modes have become standard methods of providing primary mechanical ventilatory support in critically ill patients.

Basic Principles of Airway Pressure Release Ventilation

Airway pressure release ventilation (APRV)⁶ ventilates by time-cycled switching between two pressure levels in a high-flow (or demand-valve) CPAP circuit, permitting unrestricted spontaneous breathing in any phase of the mechanical ventilator cycle (Fig. 11-1). The degree of ventilator support with APRV is determined by the duration of the two CPAP levels and the mechanically delivered tidal volume (V_T).^{6,7} V_T depends mainly on respiratory compliance and the difference between the CPAP levels. By design, changes in ventilatory demand do not alter the level of mechanical support during APRV. When spontaneous breathing is absent, APRV is not different from conventional pressure-controlled, time-cycled mechanical ventilation.^{6,7}

Figure 11-1



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Airway pressure release ventilation ventilates by time-cycled switching between a high and low continuous positive airway pressure (CPAP) level in the circuit. Consequently, unrestricted spontaneous breathing is permitted in any phase of the mechanical ventilator cycle. Change between the two CPAP levels results in a change in functional residual capacity (ΔFRC), which equals the mechanical delivered tidal volume (V_T). V_T depends mainly on respiratory compliance and resistance and the airway pressure difference (ΔP_{aw}) between the CPAP levels. Setting the time for the low (T_{low}) and the high (T_{high}) CPAP enables the adjustment of ventilator rate.

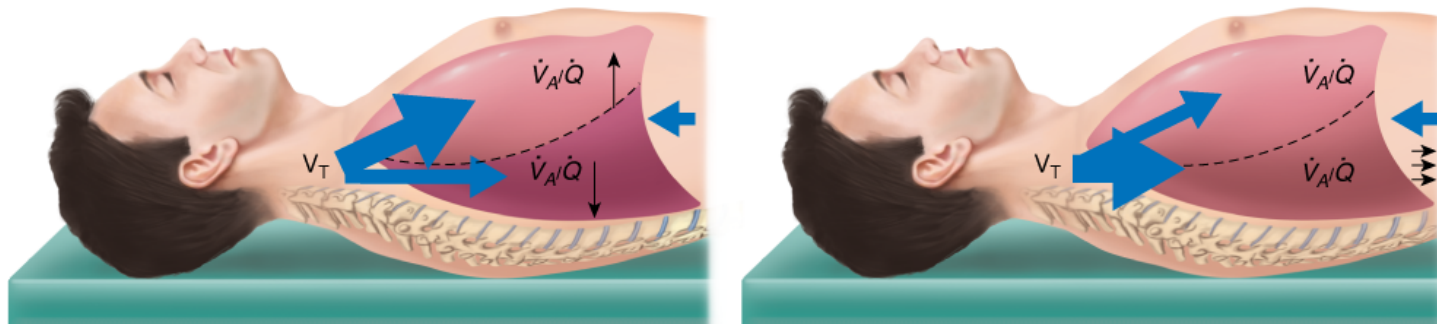
Synonyms used for APRV are biphasic positive airway pressure⁷ (BIPAP) and bilevel airway pressure (Bilevel). Biphasic positive airway pressure is identical to APRV except that no restriction is imposed on the duration of the low-CPAP level (release pressure).^{6,7} Based on the initial description, APRV keeps the duration of the low-CPAP level (release time) at 1.5 seconds or less.

Physiologic Effects

Ventilation Distributions

Radiologic studies demonstrate that ventilation is distributed differently during pure spontaneous breathing and CMV.⁸ During spontaneous breathing, the posterior muscular sections of the diaphragm move more than the anterior tendon plate.⁸ Consequently, when patients are supine, the dependent lung regions tend to be better ventilated during spontaneous breathing (Fig. 11-2). If the diaphragm is relaxed, it will be moved by the weight of the abdominal cavity and intraabdominal pressure towards the cranium; mechanical V_T will be distributed more to the anterior, nondependent, and less perfused lung regions.⁹ When compared with spontaneous breathing, the latter leads, both in patients with healthy lungs and patients with diseased lungs, to lung areas in the dorsal lung regions close to the diaphragm, being less ventilated (or atelectatic). Recent results demonstrate that the posterior muscular sections of the diaphragm move more than the anterior tendon plate when large breaths or sighs are present during spontaneous breathing.¹⁰

Figure 11-2

Mechanical ventilation**Spontaneous breathing**

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During spontaneous breathing, the posterior muscular sections of the diaphragm move more than the anterior tendon plate. Consequently, in the supine position, spontaneous ventilation is preferably directed to well-perfused, dependent lung regions. Conversely, a mechanically delivered tidal volume is directed primarily to nondependent lung areas, away from regions with maximal blood flow. Thus, spontaneous breathing contributes to better ventilation-perfusion ($\dot{V}_A/\dot{Q}$) matching.

Computed tomography (CT) of patients with acute respiratory distress syndrome (ARDS) reveals radiographic densities corresponding to alveolar collapse localized primarily in the dependent lung regions, which correlates with intrapulmonary shunting and accounts entirely for the observed arterial hypoxemia.¹¹ Formation of radiographic densities is attributed to alveolar collapse caused by superimposed pressure on the lung and a cephalad shift of the diaphragm, most evident in dependent lung areas during CMV.¹² Persisting spontaneous breathing is considered to improve the distribution of ventilation to dependent lung areas and thereby improve ventilation-perfusion ($\dot{V}_A/\dot{Q}$) matching, presumably by diaphragmatic contraction that opposes alveolar compression.¹³ This concept is supported by CT observations in anesthetized patients demonstrating that contractions of the diaphragm induced by phrenic nerve stimulation favor distribution of ventilation to dependent, well-perfused lung areas, decreasing atelectasis formation.¹⁴

Spontaneous breathing with APRV in experimentally induced lung injury is associated with less atelectasis formation on end-expiratory spiral CT of the whole lungs and on CT scans above the diaphragm (Fig. 11-3).¹⁵⁻¹⁷ Although other inspiratory muscles may contribute to improvement in aeration during spontaneous breathing, the craniocaudal gradient in aeration, aeration differences, and the marked differences in aeration in regions close to the diaphragm between APRV with and without spontaneous breathing suggest that diaphragmatic contractions play a dominant role on the observed aeration differences and improvement of end-expiratory lung volume.¹⁵⁻¹⁷ Experimental data suggest that recruitment of dependent lung areas may be caused essentially by an increase in transpulmonary pressure (P_{tp}) secondary to the decrease of pleural pressure (P_{pl}) with spontaneous breathing during APRV.¹⁸

Figure 11-3

APRV with spontaneous breathing**APRV without spontaneous breathing**

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Computed tomography of a lung region above the diaphragm at end-expiration in oleic acid-induced lung injury with and without spontaneous breathing during APRV. Atelectasis formation is reduced with spontaneous breathing.

Because the posterior muscular sections of the diaphragm contract more than the anterior tendon plate, a decrease in Ppl and the concomitant localized increase in Ptp should explain entirely the successful recruitment of atelectatic areas in the dependent lung regions adjacent to the diaphragm.^{8,10,14-17} In the apical lung regions, differences in tidal volume distribution between the dependent and nondependent lung is not significantly different with spontaneous breathing.^{15,16} Inhomogeneous distribution of tidal ventilation during CMV in patients with acute lung injury (ALI)^{19,20} may be explained by regional differences in Ptp. A cephalocaudal decrease in Ptp may be partially explained by the transmission of abdominal pressure to the thoracic cavity,¹² decreasing from base to apex. In addition, compression of lung tissue by the weight of the lungs and the heart may differ regionally.^{12,19,20} Although increase in Ptp caused by spontaneous breathing is maximal in the dependent lung areas adjacent to the diaphragm,^{8,10,14-17} it is unlikely that the resulting absolute Ptp in the dependent lung regions is higher than in cephalad lung areas in the absence of spontaneous breathing. This concept is supported by experimental findings that gas volume and aeration are not higher in dependent lung regions with spontaneous breathing than in cephalad lung regions in the absence of spontaneous breathing.¹⁷ Furthermore, CT observations in experimentally induced lung injury demonstrate that, during spontaneous breathing with APRV a large portion of the V_T is distributed to the initially collapsed dependent lung regions, resulting in less cyclical alveolar collapse.¹⁶

Pulmonary Gas Exchange

In patients with ARDS, APRV with spontaneous breathing of 10% to 30% of the total minute ventilation (V_E) accounts for an improvement in $\dot{V}_A/\dot{Q}$ matching, intrapulmonary shunting, and arterial oxygenation.¹³ These results confirm earlier investigations in animals with induced lung injury,²¹⁻²³ which demonstrated improvement in intrapulmonary shunt and arterial oxygenation during spontaneous breathing with APRV. An increase in arterial oxygenation in conjunction with greater pulmonary compliance indicates recruitment of previously nonventilated lung areas. Clinical studies in patients with ARDS show that spontaneous breathing during APRV does not necessarily lead to instant improvement in gas exchange. Instead, improvement in oxygenation continues over the 24 hours after the start of spontaneous breathing.²⁴

In patients at risk of developing ARDS, maintaining spontaneous breathing with APRV resulted in lower venous admixture and better arterial oxygenation over a period of more than 10 days as compared to CMV with subsequent weaning.²⁵ These findings are supported by randomized controlled trials and case-matched analysis.²⁶⁻²⁸

Cardiovascular Effects

Application of a ventilator breath generates an increase in airway pressure and, therefore, in intrathoracic pressure, which, in turn, reduces venous return to the heart. In normovolemic and hypovolemic patients, this action produces a reduction in right-ventricular and left-ventricular filling and results in decreased stroke volume, cardiac output, and oxygen delivery.²⁹ To normalize systemic blood flow during mechanical ventilation, intravascular volume often needs to be increased and/or the cardiovascular system needs pharmacologic support. Reducing mechanical ventilation to a level that provides adequate support for existing spontaneous breathing should help reduce the cardiovascular side effects of ventilator support.³⁰

Periodic reduction of intrathoracic pressure, achieved by maintaining spontaneous breathing during ventilator support, promotes venous return to the heart and right-ventricular and left-ventricular filling, thereby increasing cardiac output and oxygen delivery.³¹ Experimental^{21–23} and clinical^{13,25} studies show that when spontaneous breathing during APRV achieves 10% to 40% of total V_E , with no change in V_E or airway pressure limits, the cardiac index increases. A simultaneous rise in right-ventricular end-diastolic volume during spontaneous breathing with APRV indicates improved venous return to the heart.¹³ In addition, outflow from the right ventricle, which depends mainly on lung volume (the major determinant of pulmonary vascular resistance), may benefit from decrease in intrathoracic pressure during APRV.¹³

Patients with left-ventricular dysfunction may not benefit from either the augmentation of venous return to the heart or the increase in left-ventricular afterload that occurs with the lowering of intrathoracic pressure. Thus, switching abruptly from CMV to pressure-support ventilation (PSV) with a simultaneous reduction in airway pressure can cause further decompensation in patients with existing cardiac insufficiency.³² Räsänen et al³³ showed a need for adequate ventilatory support and CPAP levels in patients with respiratory and cardiogenic failure. Provided that spontaneous breathing receives adequate support and that satisfactory CPAP levels are applied, maintaining spontaneous breathing during APRV should not be a disadvantage and is not per se contraindicated in patients with ventricular dysfunction.³³

Oxygen Supply and Demand

An increase in cardiac index and arterial oxygenation (Pa_{O_2}) during APRV improves the relationship between tissue oxygen supply and demand because oxygen consumption remains unchanged despite the work of spontaneous breathing. In accordance with previous experimental^{21,22} and clinical findings,¹³ total oxygen consumption is not measurably altered by adequately supported spontaneous breathing in patients with low lung compliance. An increase in oxygen delivery with unchanged oxygen consumption indicates an improved tissue oxygen supply and demand balance, as reflected by a decrease in oxygen extraction ratio and higher mixed venous partial pressure of oxygen.

Organ Perfusion

By reducing cardiac index and venous return to the heart, CMV can have a negative effect on the perfusion and functioning of extrathoracic organ systems. An increase in venous return and cardiac index, secondary to the periodic fall in intrathoracic pressure during spontaneous inspiration, should significantly improve organ perfusion and function during partial ventilator support. In patients with ARDS during spontaneous breathing with APRV, renal perfusion and glomerular filtration rate improve.³⁴ Using the colored microsphere technique, Hering et al^{35,36} observed (in an experimental model) that spontaneous breathing during APRV improves systemic and mucosal-submucosal blood flow in the gastrointestinal tract as compared to CMV (with and without permissive hypercapnia); hepatic arterial blood flow remained essentially unchanged. In the absence of an increased intracranial pressure, regional cerebral and spinal cord blood flows were higher with spontaneous breathing.³⁷

Rationale, Advantages, and Limitations

Based on physiologic observations, APRV is advantageous for recruiting atelectasis adjacent to the diaphragm, thereby improving pulmonary gas exchange in patients with ALI, ARDS, and atelectasis after major surgery. Because increase in Ptp is localized to the areas near the diaphragm and is caused by a decrease in pleural pressure, the concomitant decrease in intrathoracic pressure contributes to improved cardiovascular function. Areas of atelectasis not adjacent to the diaphragm may not be successfully recruited by spontaneous breathing during APRV.

To enable spontaneous breathing, lower levels of sedation (Ramsay Score of 2 to 3) are required. Less sedation helps to reduce the dosages of vasopressor and inotropic agents, while maintaining cardiovascular function stability. In addition, less sedation reduces the duration of ventilator support.²⁵ The use of APRV, however, has to be limited to patients who do not require deep sedation for management of their underlying disease (e.g., cerebral edema with increased intracranial pressure).

Two periods during the APRV cycle are particularly vulnerable to patient-ventilator asynchrony. When airway pressure release occurs during spontaneous inspiration and when restoration of CPAP occurs during spontaneous expiration, ventilation may be impaired because spontaneous and ventilator efforts oppose each other. Rarely, a reduction in ventilatory efficiency, indicated by a decrease in alveolar ventilation and an increase in

work of breathing, may result from temporary asynchrony. Synchronized APRV and optimizing ventilator settings and sedation may be required in this rare event.³⁸

As a concept, APRV does not provide breath-to-breath assistance to spontaneous inspiration. Previous investigations demonstrated that separation from mechanical ventilation in difficult-to-wean patients may be prolonged with use of intermittent mandatory ventilation (IMV) and may be expedited with breath-to-breath assistance to inspiratory efforts during PSV.⁴ Thus, APRV is not expected to be an advantage in difficult-to-wean patients.

Indications and Contraindications

Indications

Based on clinical^{13,25,26–28,39} and experimental^{15–17,21,22} data, APRV is indicated in patients with ALI, ARDS, and atelectasis after major surgery. APRV recruits atelectasis adjacent to the diaphragm, thereby restoring pulmonary gas exchange, and improves cardiovascular and extrathoracic organ function in patients with ALI or ARDS, and atelectasis after major surgery.

Contraindications

Because lower levels of sedation (Ramsay Score of 2 to 3) are used to enable spontaneous breathing, APRV should not be used in patients who require deep sedation for management of their underlying disease (e.g., cerebral edema with increased intracranial pressure).

To date, no data are available on use of APRV in patients with obstructive lung disease. Theoretically, use of a short release time should not be beneficial in patients with obstructive lung disease who have prolonged expiratory time constants. Currently, use of APRV is not supported in these patients by clinical research.

Likewise, use of APRV has not been investigated in patients with neuromuscular disease, and is not supported by any evidence.

Comparison with Other Modes

Airway Pressure Release Ventilation versus Pressure-Support Ventilation

APRV and PSV were compared in twenty-four patients with ALI and/or ARDS using equal V_E or airway pressure limits. Because insufflation during PSV is flow-cycled, alveolar end-inspiratory pressure may not reach the preset level. Thus, in patients with reduced lung compliance, equal airway pressure limits achieve a lower V_T during PSV as compared with APRV. Consequently, a compensatory increase in respiratory rate is required during PSV to maintain alveolar ventilation. To deliver APRV and PSV with comparable V_T at an acceptable respiratory rate, the pressure level has to be increased during PSV.¹³

In contrast to spontaneous breathing with APRV, assisted inspiration with PSV did not produce significant improvement in intrapulmonary shunt, gas exchange or cardiac output when compared with CMV.¹³ Apparently, the spontaneous contribution to a mechanically assisted breath was not sufficient to counteract the $\dot{V}_A/\dot{Q}$ maldistribution and cardiovascular depression caused by positive-pressure ventilation. A possible explanation might be that inspiration is terminated by the decrease in inspiratory gas flow during PSV, which may reduce ventilation in areas of the lung that have a slow time constant.

Airway Pressure Release Ventilation versus Intermittent Mandatory Ventilation

In a randomized multicenter trial in fifty-two patients with ALI, APRV with lower peak airway pressures resulted in a better alveolar ventilation and equal arterial oxygenation as compared with IMV.²⁶ A similar trial in fifty-eight patients with ALI supports these findings, but did not show a difference in mortality.⁴⁰ In eight patients recovering from open-heart surgery, APRV provided adequate ventilation with lower airway pressures and less dead-space ventilation than did IMV or PSV.⁴¹ Arterial oxygenation was not different between the modalities.

Airway Pressure Release Ventilation versus Assist-Control Ventilation

A comparison of 234 patients ventilated with APRV with 234 patients ventilated with assist-control ventilation in a case-matched analysis revealed no differences in days of mechanical ventilation or weaning, rate of reintubation, length of stay in the intensive care unit or the hospital, and mortality in the intensive care unit or the hospital.²⁸ In this study, APRV was used in almost all disease states that lead to acute respiratory failure. The doses and duration of sedatives and analgesics were not different between the groups and the amount of spontaneous ventilation during APRV was not reported.

Likewise, a randomized controlled trial involving sixty-two trauma patients showed no differences between APRV and assist-control ventilation for ventilator days, length of stay in the intensive care unit, pneumothorax, ventilator-associated pneumonia, or mortality.²⁷

Variation in Delivery among Ventilator Brands

Synchronized Airway Pressure Release Ventilation

Asynchronous interferences between spontaneous and mechanical ventilation may increase the work of breathing and reduce the effective support during APRV.³⁸ Synchronization of the switching between the two CPAP levels to spontaneous inspiration or expiration has been incorporated in all commercially available demand-valve APRV circuits to avoid asynchronous interferences between spontaneous and mechanical breaths. A trigger window of 0.25 seconds is usually used to enable synchronization of the switching between the two CPAP levels and spontaneous breathing efforts. Bench-model data indicate that the synchronization of spontaneous inspiration with the switch to the high CPAP level, but not the synchronization of spontaneous expiration synchronized with pressure release, may be beneficial. Patient data on the advantage of synchronized APRV are lacking currently. Because patient-triggered mechanical cycles during IMV are not advantageous for patients, there is no reason why this should be different for APRV. Synchronization during APRV may produce inconstant times for the high and low CPAP. Synchronization of APRV is switched off in the APRV mode of Dräger Evita IV, XL, and V500 ventilators. In the Servo I, the Bennett 840, Hamilton G5, and the Viasys Vela and Avea ventilators, APRV is synchronized with spontaneous ventilation.

Modifications of Airway Pressure Release Ventilation

Most commercially available ventilators offer hybrid modes of ventilation such as APRV + PSV and APRV + automatic tube compensation (Table 11-1). Very few of these combinations have been shown to benefit patients.⁴² It is doubtful that simply combining different modalities will achieve an addition of their positive effects.^{43,44} Indeed, it is possible that proven physiologic benefit of one modality might be minimized or even neutralized when it is by combined with another mode.

Table 11-1: Modifications of Airway Pressure Release Ventilation

• Synchronized APRV ◦ Change between CPAP levels is synchronized with spontaneous breathing ◦ Advantage of synchronization is not proven • Intermittent mandatory pressure release ventilation ◦ Spontaneous breathing on the high CPAP level is assisted with PSV ◦ Advantage is not supported by data • APRV + PSV ◦ Spontaneous breathing on the low CPAP level is assisted with PSV ◦ Advantage is not supported by data • APRV + ATC ◦ Spontaneous breathing on both CPAP levels is assisted with ATC ◦ May reduce work of breathing in selected patients without deteriorating gas exchange
Abbreviations: <i>APRV</i> , Airway pressure release ventilation; <i>ATC</i> , automatic tube compensation; <i>CPAP</i> , continuous positive airway pressure; <i>PSV</i> , pressure-support ventilation.

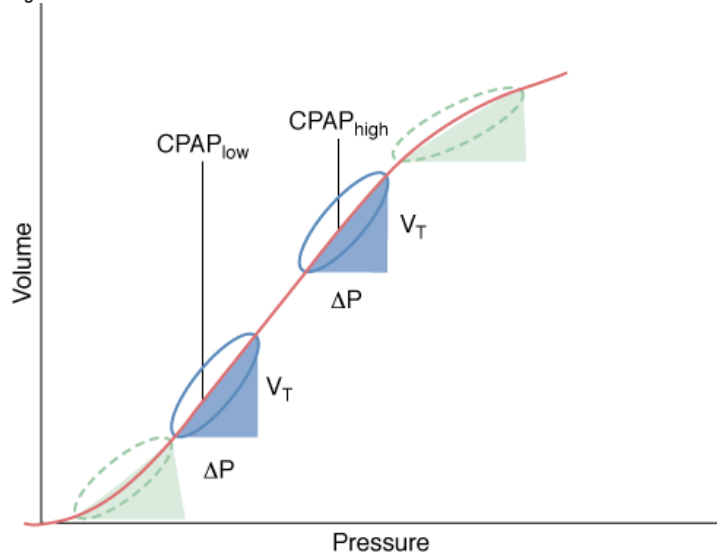
Adjustments at the Bedside

Setting Ventilation Pressures and Tidal Volumes during Airway Pressure Release Ventilation

Mechanical ventilation with positive end-expiratory pressure (PEEP) titrated above the lower inflection pressure of a static pressure–volume curve and a low VT is thought to prevent tidal alveolar collapse at end-expiration and overdistension of lung units at end-inspiration in patients with ARDS.⁴⁵ This lung-protective strategy causes improvement in lung compliance, venous admixture, and PaO2 without causing cardiovascular impairment in ARDS.⁴⁵ Mechanical ventilation using VT of not more than 6 mL/kg (ideal body weight) has been shown to result in a better outcome when compared with a VT of 12 mL/kg (ideal body weight) in patients with ARDS.^{45,46} Based on these results, CPAP levels during APRV should be titrated to prevent

end-expiratory alveolar collapse and tidal alveolar overdistension.^{45,46} When CPAP levels during APRV were adjusted in patients with ARDS according to a lung-protective strategy, occurrence of spontaneous breathing led to improved cardiorespiratory function without affecting total oxygen consumption secondary to the work of breathing.¹³ Moreover, pulmonary compliance should be greatest in this range of airway pressures and, thus, a small change in transpulmonary pressure achieves normal tidal breathing with minimal elastic work of breathing (Fig. 11-4).⁴⁷ Because APRV does not provide assistance on every inspiratory effort, the CPAP levels need to be carefully adjusted to achieve efficient spontaneous ventilation with minimal work of breathing.

Figure 11-4



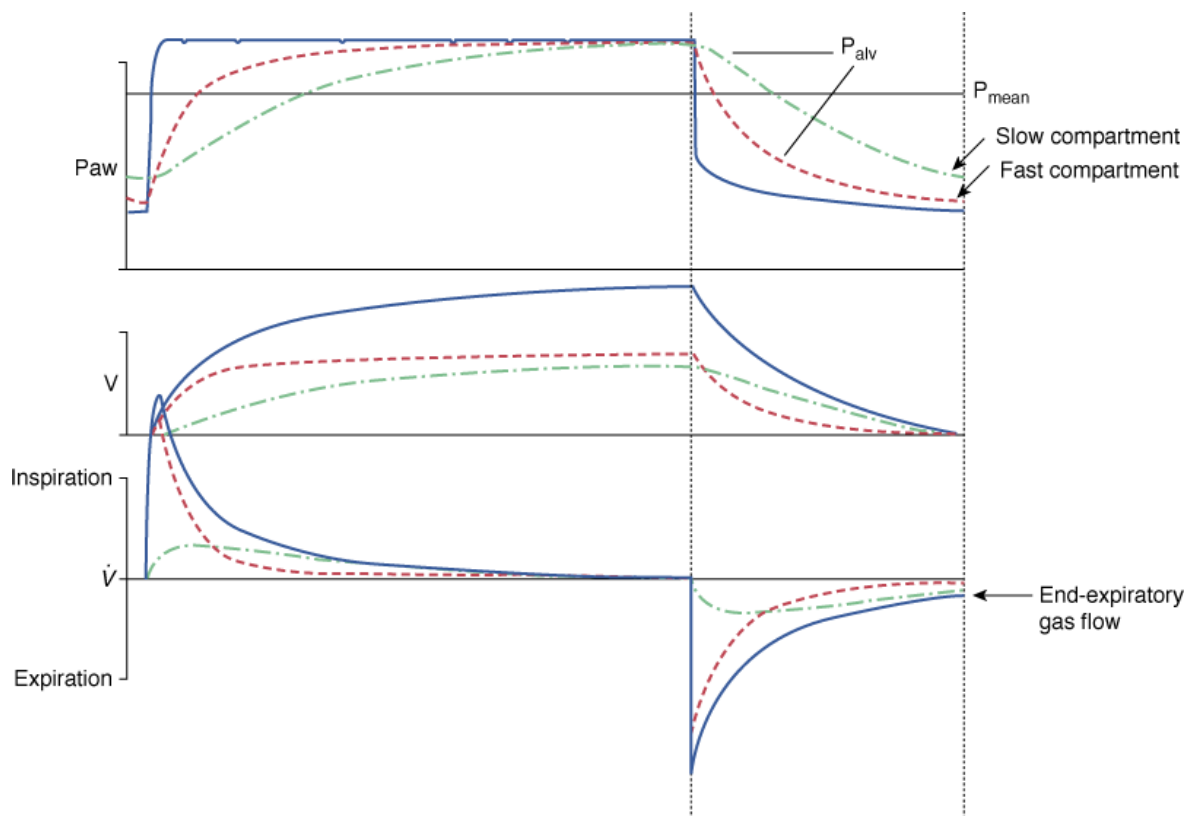
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During airway pressure-release ventilation, both CPAP levels should be titrated to achieve the greatest compliance. Thus, a small change in transpulmonary pressure (ΔP) achieves normal tidal breathing (V_T), while elastic work of breathing (*shaded area*) is minimal. If the CPAP levels are too high or too low (*dashed lines*), elastic work of breathing will be increased unnecessarily (*shaded areas*).

Setting Times during Airway Pressure Release Ventilation

The duration of the high-CPAP level needs to allow at least complete inflation of the lungs, as indicated by an end-inspiratory phase of no flow when spontaneous breathing is absent. Spontaneous breathing occurs normally on the high-CPAP level. Thus, duration of the high-CPAP level should be adjusted so that it is long enough to allow spontaneous breathing. If the release time is shorter than four times the time constant of the lungs (τ = compliance $\times$ resistance), alveolar pressure will not equilibrate at the low-CPAP level, and intrinsic PEEP (PEEPi) will result.^{48,49} Incomplete expiration and the likelihood of PEEPi is indicated by gas flow at end-expiration (Fig. 11-5). In the presence of PEEPi, alveolar pressure amplitude will be reduced; consequently, alveolar ventilation decreases and partial pressure of carbon dioxide increases. To date, data do not indicate that PEEPi is superior to external PEEP in preventing derecruitment of the lungs. Thus, duration of the low CPAP level should be adjusted to allow complete expiration to resting lung volume.

Figure 11-5



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Computer simulation of airway pressure (P_{aw}), volume (V), and gas flow ($\dot{V}$) for the respiratory system, and both a fast and slow lung compartment with a short release time. Expiration in the slow compartment is not completed at end-expiration; consequently, gas flow is still present at end-expiration and associated with intrinsic PEEP.

Other Concepts of Setting Airway Pressure Release Ventilation

Other approaches use high-CPAP levels, which are briefly released to near-ambient pressure during APRV. Depending on the time constant of the lungs, brief release times may cause PEEPi. Clinical studies, however, demonstrate that external PEEP is superior to PEEPi in restoring gas exchange in patients with ALI. Not surprisingly, in patients with ARDS, Cane et al²⁶ observed an increase in atelectasis upon briefly releasing high-CPAP levels to near-ambient pressure during APRV. Recent trials and case series based on this method of adjusting APRV reported better oxygenation when APRV was used as a rescue strategy in severely hypoxemic patients with ARDS.^{27,50,51}

Troubleshooting

Analgesia and Sedation during Airway Pressure Release Ventilation

Analgesia and sedation are used not only to ensure satisfactory pain relief and anxiolysis, but also to help the patient adapt to mechanical ventilation.⁵² The additional use of neuromuscular paralysis is controversial.⁵³ The level of analgesia and sedation required during CMV is equivalent to a Ramsay score higher than 5; that is, a deeply sedated patient unable to respond when spoken to and who has no sensation of pain. During APRV, a Ramsay score of 2 to 3 can be targeted; that is, an awake, responsive, and cooperative patient. In nearly 600 postcardiac surgery patients,³⁹ and in patients with multiple injuries,²⁵ maintaining spontaneous breathing during APRV led to less consumption of analgesics and sedatives as compared with initial use of CMV followed by weaning with partial ventilator support. Higher doses of analgesics and sedatives used in patients managed with CMV are associated with the use of higher doses of vasopressors and inotropic agents to maintain stable cardiovascular function,²⁵ and have been suggested to produce a higher incidence of delirium, longer duration of mechanical ventilation and stay in the intensive care unit, and mortality.⁵⁴

Important Unknowns

Although prospective randomized trials have found improved cardiopulmonary function in the absence of an increased mortality or longer duration of mechanical ventilation with the use of APRV, concerns have been raised as to whether an increased transpulmonary pressure consequent to

spontaneous breathing efforts can contribute to ventilator-induced lung injury. In addition, the effect of lower levels of sedation during APRV on the incidence of delirium and mortality has yet to be investigated.

The Future

Randomized controlled studies and case-matched analysis have demonstrated that early spontaneous breathing with APRV in patients with ALI/ARDS and in patients with pulmonary dysfunction after trauma and major surgery leads to improved arterial oxygenation and cardiovascular function. Larger randomized multicenter trials are needed to test the validity of these results in critically ill patients.

Summary and Conclusion

Based on available data, it is suggested that spontaneous breathing during ventilator support does not need to be suppressed, even in patients with severe pulmonary dysfunction, if there is no contraindications (e.g., increased intracranial pressure). Improvement in pulmonary gas exchange, systemic blood flow, and [oxygen](#) supply to the tissue with less consumption of analgesics and sedatives has been observed even in severe ARDS. Maintaining spontaneous breathing with APRV has not yet been shown to significantly change outcome. [CMV](#) followed by weaning with partial ventilator support is the standard in ventilation therapy, but the place of spontaneous breathing with APRV should be reconsidered in view of available data. Today's standard practice should be to maintain spontaneous breathing as early as possible during ventilator support and to continuously adapt the level of support according to the individual needs of the patient.

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